

Predictive N_T — Six-Component COSMO-SAC-2010

RESEARCH DIAGNOSTIC — NOT ACCEPTED — NOT RELEASE ELIGIBLE

Executive verdict

The calculation completed successfully for stage counts 1 through 10. No trial simultaneously satisfied all numerical, thermodynamic, component-balance, and saved product-target gates. Predictive N_T and established theoretical stages therefore remain unassigned.

Job status	COMPLETED — no runtime error
Trials completed	10 / 10
Diagnostic Predictive N_T	Not assigned
Established theoretical stages	Not established
Sulfur prediction	NOT CALCULABLE
Pilot validated	No
Calibration required	Yes
Release eligible	No

ENGINEERING INTERPRETATION

“Not accepted” is a scientific qualification verdict, not a software failure. The run provides diagnostic evidence showing where recovery, solvent carryover, PA, stability, closure, or saved product-target gates fail.

Completed 31/8/2026, 10:37:05 am IST

Frozen job: 1acd1a68-51a0-4877-94fc-0f199eb9f467

1. Frozen Stage 1 design basis

Saved authority used without live-input substitution

Project reference	111
RRBO grade	SN300
Feed rate	4000 L/h
Operating temperature	30 °C
Operating pressure	2.0
Phase configuration	nmp-continuous-rrbo-dispersed
Solvent / oil mass ratio	0.75
Maximum stage search	10
Feed sulfur	3500 ppm
Sulfur target	1000 ppm — NOT CALCULABLE

Feed composition — wt%

Basis	SAT	MONO	DI	POLY	PA	NMP
Feed	85	7	4	2	2	0

Saved product targets

Minimum NMP-free RRBO recovery	95%
Maximum NMP in total raffinate	0.5 wt%
Minimum saturates, NMP-free	90 wt%
Maximum total aromatics, NMP-free	5 wt%
Maximum PA, NMP-free	0.1 wt%

MODELED SOLVENT BASIS

Fresh solvent is pure modeled NMP at the saved S/O ratio. Commercial purity and water specifications are audit-only and do not add components.

2. Ten-trial comparison dashboard

Frozen completed result; composition metrics are mass percentages

Stages	Solver	Numerical	Recovery %	NMP wt%	SAT wt%	Aromatics wt%	PA wt%	Targets	Qualified
1	YES	FAIL	87.69	7.57	91.68	7.57	0.76	FAIL	FAIL
2	YES	FAIL	84.44	7.16	94.48	5.19	0.32	FAIL	FAIL
3	YES	FAIL	82.79	6.95	95.97	3.88	0.14	FAIL	FAIL
4	YES	FAIL	81.81	6.82	96.87	3.07	0.06	FAIL	FAIL
5	YES	FAIL	81.18	6.74	97.46	2.51	0.03	FAIL	FAIL
6	YES	FAIL	80.75	6.68	97.87	2.12	0.01	FAIL	FAIL
7	YES	FAIL	80.43	6.64	98.17	1.83	0.01	FAIL	FAIL
8	NO	FAIL	80.25	6.62	98.29	1.7	0.01	FAIL	FAIL
9	NO	FAIL	81.55	6.72	96.96	2.56	0.49	FAIL	FAIL
10	NO	FAIL	81.3	6.71	97.04	2.48	0.47	FAIL	FAIL

Limits: recovery "e95%; NMP "d0.5 wt%; SAT "e90 wt%; total aromatics "d5 wt%; PA "d0.1 wt%. Sulfur is not calculable.

RESULT: NO QUALIFIED TRIAL — PREDICTIVE N_T NOT ASSIGNED

3. Product-target compliance — trials 1 to 5

Each card shows calculated value / saved target

1 STAGE			NOT QUALIFIED
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt	
FAIL 87.69 / 95	FAIL 7.57 / 0.5	PASS 91.68 / 90	
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm	
FAIL 7.57 / 5	FAIL 0.76 / 0.1	NOT CALCULABLE — / 1000	
2 STAGES			NOT QUALIFIED
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt	
FAIL 84.44 / 95	FAIL 7.16 / 0.5	PASS 94.48 / 90	
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm	
FAIL 5.19 / 5	FAIL 0.32 / 0.1	NOT CALCULABLE — / 1000	
3 STAGES			NOT QUALIFIED
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt	
FAIL 82.79 / 95	FAIL 6.95 / 0.5	PASS 95.97 / 90	
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm	
PASS 3.88 / 5	FAIL 0.14 / 0.1	NOT CALCULABLE — / 1000	
4 STAGES			NOT QUALIFIED
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt	
FAIL 81.81 / 95	FAIL 6.82 / 0.5	PASS 96.87 / 90	
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm	
PASS 3.07 / 5	PASS 0.06 / 0.1	NOT CALCULABLE — / 1000	
5 STAGES			NOT QUALIFIED
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt	
FAIL 81.18 / 95	FAIL 6.74 / 0.5	PASS 97.46 / 90	
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm	
PASS 2.51 / 5	PASS 0.03 / 0.1	NOT CALCULABLE — / 1000	

3. Product-target compliance — trials 6 to 10

Each card shows calculated value / saved target

6 STAGES			NOT QUALIFIED		
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt			
FAIL 80.75 / 95	FAIL 6.68 / 0.5	PASS 97.87 / 90			
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm			
PASS 2.12 / 5	PASS 0.01 / 0.1	NOT CALCULABLE — / 1000			

7 STAGES			NOT QUALIFIED		
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt			
FAIL 80.43 / 95	FAIL 6.64 / 0.5	PASS 98.17 / 90			
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm			
PASS 1.83 / 5	PASS 0.01 / 0.1	NOT CALCULABLE — / 1000			

8 STAGES			NOT QUALIFIED		
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt			
FAIL 80.25 / 95	FAIL 6.62 / 0.5	PASS 98.29 / 90			
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm			
PASS 1.7 / 5	PASS 0.01 / 0.1	NOT CALCULABLE — / 1000			

9 STAGES			NOT QUALIFIED		
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt			
FAIL 81.55 / 95	FAIL 6.72 / 0.5	PASS 96.96 / 90			
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm			
PASS 2.56 / 5	FAIL 0.49 / 0.1	NOT CALCULABLE — / 1000			

10 STAGES			NOT QUALIFIED		
minimum Nmp Free Recovery Pct	maximum Nmp Raffinate Wt	minimum Raffinate Saturates Wt			
FAIL 81.3 / 95	FAIL 6.71 / 0.5	PASS 97.04 / 90			
maximum Raffinate Total Aromatics Wt	maximum Raffinate Polar Aromatics Wt	target Raffinate Sulfur Ppm			
PASS 2.48 / 5	FAIL 0.47 / 0.1	NOT CALCULABLE — / 1000			

4. Boundary-stream composition

Representative stage counts; all component values are wt%

1 stage

Stream	SAT	MONO	DI	POLY	PA	NMP	Mass
oil Feed	85	7	4	2	2	0	100
fresh Nmp	0	0	0	0	0	100	75
final Raffinate	84.73	3.84	2.15	1.01	0.7	7.57	94.87
final Extract	5.75	4.19	2.45	1.3	1.67	84.64	80.13

5 stages

Stream	SAT	MONO	DI	POLY	PA	NMP	Mass
oil Feed	85	7	4	2	2	0	100
fresh Nmp	0	0	0	0	0	100	75
final Raffinate	90.89	1.36	0.72	0.26	0.03	6.74	87.05
final Extract	6.69	6.61	3.84	2.02	2.25	78.61	87.95

10 stages

Stream	SAT	MONO	DI	POLY	PA	NMP	Mass
oil Feed	85	7	4	2	2	0	100
fresh Nmp	0	0	0	0	0	100	75
final Raffinate	90.54	1.02	0.74	0.56	0.44	6.71	87.15
final Extract	6.89	7.42	3.88	1.84	2.28	77.69	89

5. Numerical diagnostics

Scientific solver completion is distinct from trial qualification

Stages	Max scaled residual	Mole balance	Mass balance	Multistart difference	Gate
1	2.65e-14	2.01e-16	3.43e-14	4.39e-11	FAIL
2	6.38e-14	7.63e-17	1.30e-14	0	FAIL
3	2.78e-14	4.22e-16	1.71e-13	0	FAIL
4	4.40e-14	1.39e-17	2.36e-15	0	FAIL
5	1.06e-13	2.61e-13	1.06e-10	0	FAIL
6	8.62e-14	1.48e-15	6.02e-13	0	FAIL
7	2.86e-13	7.58e-13	3.07e-10	0	FAIL
8	9.00e-5	0	0.08	6.57e-5	FAIL
9	0	0	0.47	0	FAIL
10	0	0	0.5	0	FAIL

Qualification blockers

- 1: POST_SPLIT_TPD_STABILITY_FAILED
- 2: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 3: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 4: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 5: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 6: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 7: MULTISTART_BRANCH_REPRODUCTION_FAILED; MULTISTART_SECONDARY_CLOSURE_FAILED
- 8: COUPLED_SOLVER_CLOSURE_FAILED; MULTISTART_BRANCH_REPRODUCTION_FAILED
- 9: COUPLED_SOLVER_CLOSURE_FAILED; MULTISTART_BRANCH_REPRODUCTION_FAILED
- 10: COUPLED_SOLVER_CLOSURE_FAILED; MULTISTART_BRANCH_REPRODUCTION_FAILED

6. Governance and provenance

Immutable identifiers for the frozen completed snapshot

CONTROLLED RESEARCH OUTPUT — CALIBRATION REQUIRED

Thermodynamic classification	SIX_COMPONENT_COSMO_SAC_2010_RESEARCH_DIAGNOSTIC
Model SHA-256	9666ff67194102ca9c3dc139695bead8ff133fbfe24c2d786ce1deca49456196
Engine SHA-256	5d9ab380d8dcadaaed3a8ada30dec216700f8215f8649549491c456a9672057d
Input SHA-256	93486a655df80b5d6c61d2e3400fbe8b9610697ab50edab0bf893424199197df
Stage 1 authority SHA-256	—
Six-component basis SHA-256	7683442c0419659d83320cfa4433fe8cf298f285f596c4676cf39bb5042838fc
Six-component binding SHA-256	d71161802f9294719416b209739f7918c27cda87a716162fe9308d948271049a
Job ID	1acd1a68-51a0-4877-94fc-0f199eb9f467

Mandatory limitations

- Sulfur removal is NOT CALCULABLE from this aromatic-transfer model.
- The PA representative is non-sulfur-bearing and is not a sulfur surrogate.
- No trial is pilot validated, accepted for design, or release eligible.
- No theoretical-stage count is established by this run.
- Calibration and direct matching NMP/RRBO evidence remain required.

DOCUMENT CONTROL

Generated only from the persisted completed job snapshot. No live design inputs were read and no scientific calculation was rerun for this report.